

# Ganesh Gupta

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## EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Owned **backend/systems** work end to end: a pluggable **distributed executor** across **Docker** and **AWS Lambda** for AI agent workloads at **millions of daily transactions**.
- Architected **multi-agent AI** systems with **LangGraph**, **CrewAI**, and the **OpenAI Agents SDK**, taking them from prototype to production at **millions of daily transactions** and **99%+ uptime**.
- Designed **agent state management** with **Redis checkpointing** so interrupted multi-step workflows resume from the last committed step, making long tool-call chains **reliable**.
- Built **Prometheus/Grafana** observability and **autoscaling** for **vLLM** inference serving, correlating compute and network telemetry to catch regressions early.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Owned **ML infrastructure** end to end, from model deployment to evaluation and debugging, cutting **p99 latency 420ms to 140ms** across **50K+ daily requests**.
- Optimized LLM serving with **CUDA/Triton** kernel work, **speculative decoding**, and **TensorRT-LLM** alongside **vLLM** on **Kubernetes**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** for **1M+ users** on **AWS** (EC2, RDS, SQS), sustaining **99.9% uptime** over four years.
- Profiled compute and memory bottlenecks** across **Python/C++** services and optimized batching and **concurrency**, raising throughput **2.4x** and cutting memory **27%**.

## PROJECTS

Distributed KV Store

[github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Raft-consensus** distributed key-value store with **linearizable reads**, leader election, and **log compaction**.

RootCause AI

[github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- Multi-agent** incident-diagnosis system with **tool-use** chains that collects evidence from logs and metrics and synthesizes root-cause reports.

Watch-your-LLM

[github.com/ganeshhgupta/watch-your-llm](https://github.com/ganeshhgupta/watch-your-llm)

- LLM evaluation** platform with rubric-based scoring, drift detection, and latency tracking across model versions; **FastAPI**, **PostgreSQL**, regression alerting.

## EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

## SKILLS

**Languages:** Python, Go, C++, TypeScript, SQL

**Systems:** Distributed Systems, Agent Infrastructure, Low Latency, Concurrency, Redis, PostgreSQL

**AI:** LLM Serving, vLLM, Agents, MCP

**Cloud & DevOps:** AWS, GCP, Kubernetes, Docker, Terraform, CI/CD, Prometheus, Grafana

**Fundamentals:** Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, HTML, CSS

**Certifications:** AWS Certified Developer Associate, Google Cloud Professional Machine Learning Engineer